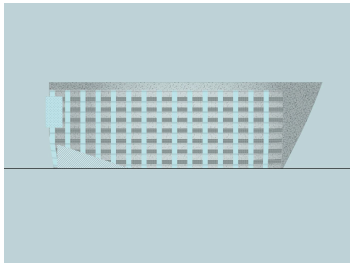
PORTFOLIO · ARCHITECTURE

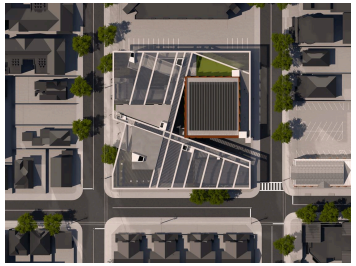
# JOHN NORTHRUP

## Designer & Digital Fabrication Specialist | Computational Workflows & Physical Systems

I am a designer and technical fabricator specializing in computational modeling, digital workflows, and physical construction. Grounded in hands-on material fabrication—from fine joinery to structural welding—and technical spatial design, my work bridges complex digital systems with buildable, physical space. I focus on technical design development, project management, and translating complex digital models into precise physical outcomes.







JCNORTHRUP@GMAIL.COM JCNORTHRUP.COM

ARCHITECTURE


| | | | |
|---|---|---|---|
| 01 | DESIGN SCHOOL IN MEXICO CITY | SP25, 2GB STUDIO, WITH NATE SMITH | 03 |
| 02 | LITTLE TOKYO THEATER | FA23, DS 1100, 1GA STUDIO | 06 |
| 03 | LA DANCE SHED | FA24, 2GA STUDIO, WITH MILAN SLEDGE | 13 |

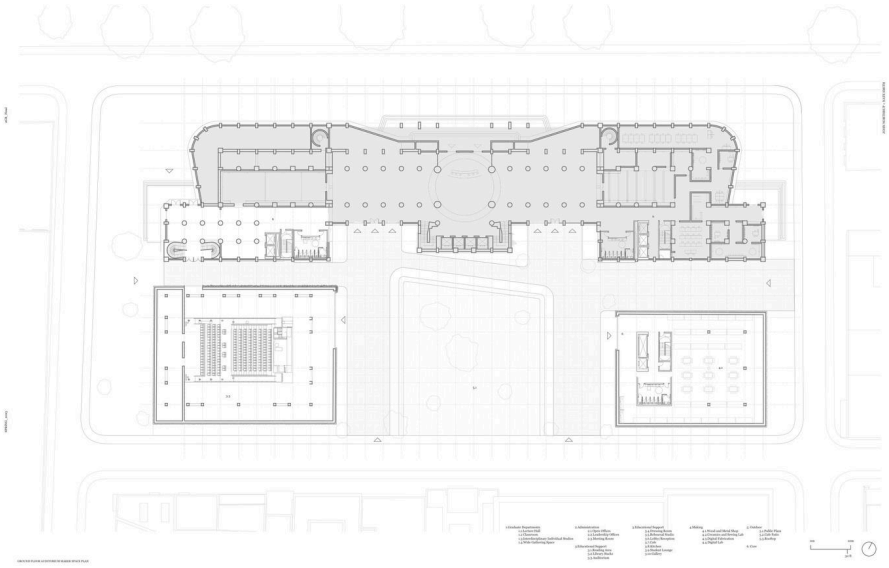


01

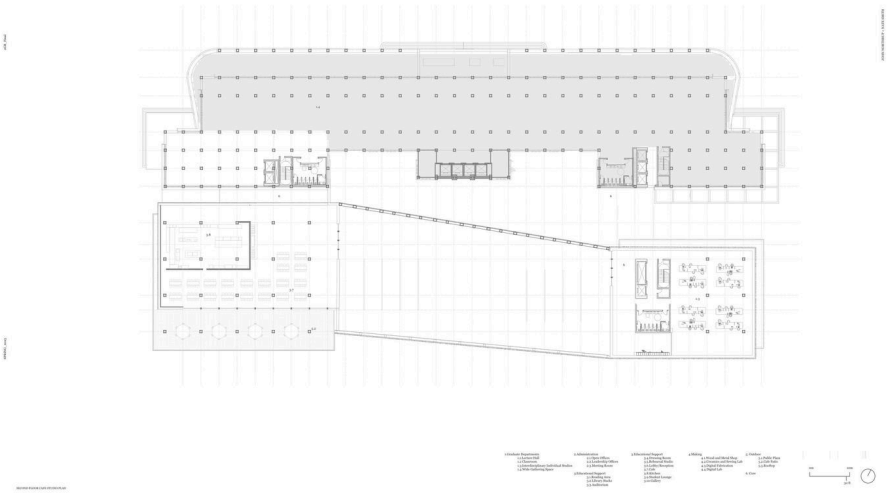
# DESIGN SCHOOL IN MEXICO CITY

SP25, 2GB STUDIO, WITH NATE SMITH
ADVISOR · ZEINA KOREITEM

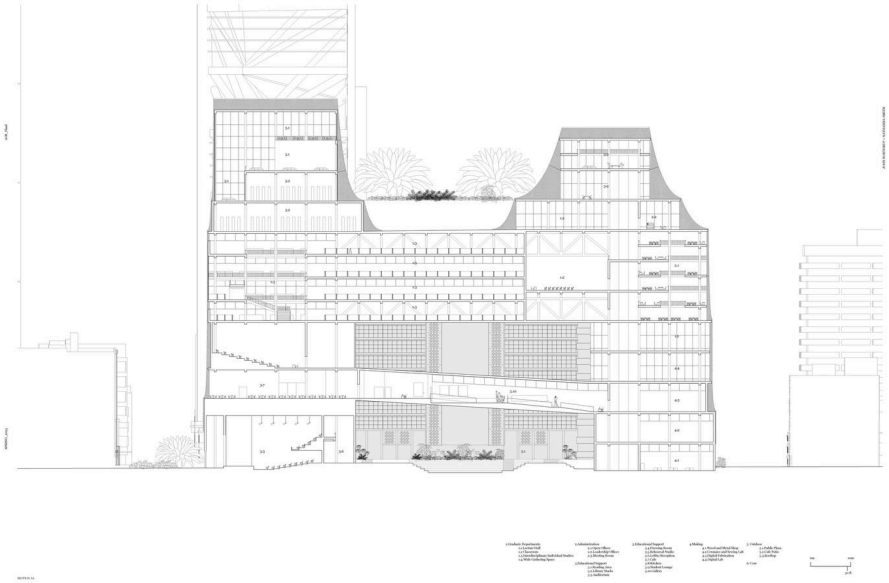
An addition to the IMSS headquarters, Mexico's social security bureau, in Mexico City: a design school that plugs into the existing office building without major changes to its structure. On our studio trip we read the city as a palimpsest, built in layers, with new buildings and materials laid directly against the ones before them. The proposal continues that reading. Studios, classrooms, a gallery, a café and a library stack as blocks on one another and on the IMSS building, a public plaza opens at street level, and a mesh facade drapes over the whole: one more layer of the city.



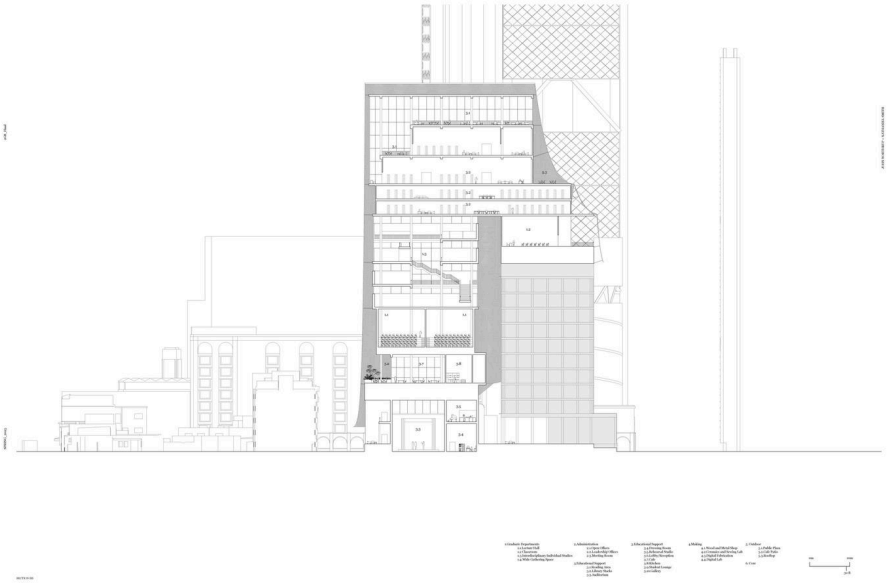
DRAWING | GROUND FLOOR PLAN AND PUBLIC PLAZA



DRAWING | SECOND FLOOR, CAFÉ AND STUDIOS



DRAWING | LONG SECTION AA



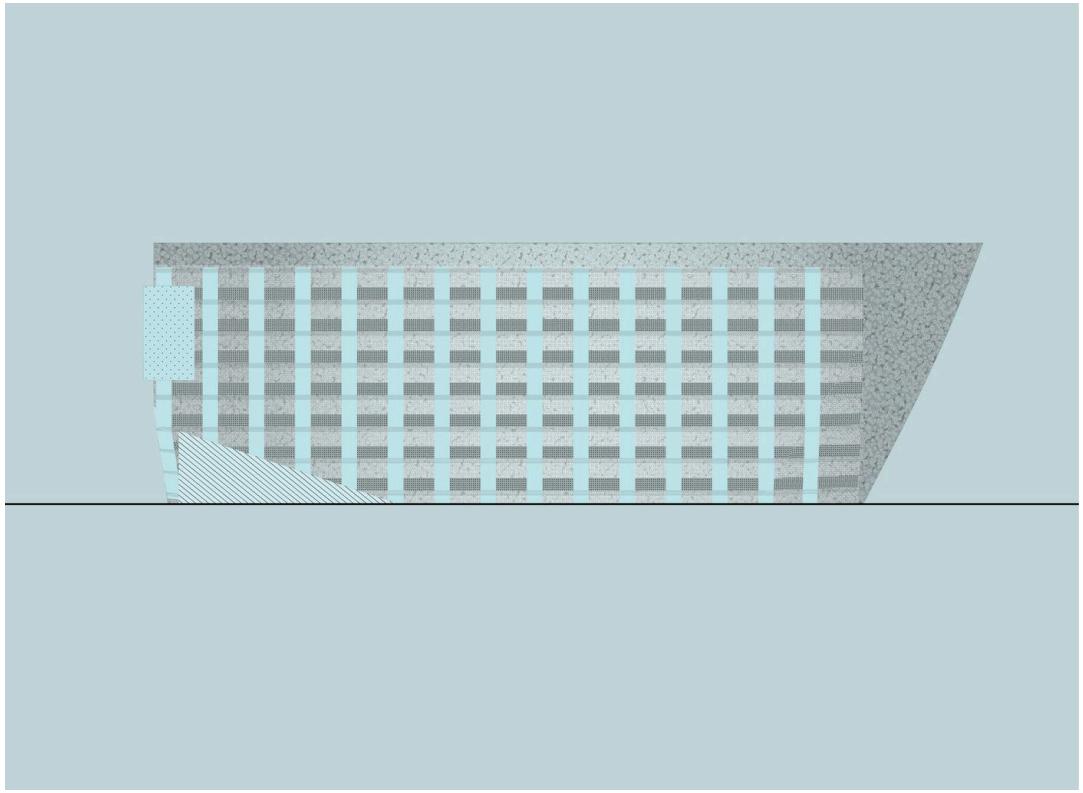
DRAWING | SHORT SECTION BB



MODEL | FINAL MODEL, MESH REMOVED



MODEL | BRICK-CLAD BLOCKS AND TRUSS BRIDGE



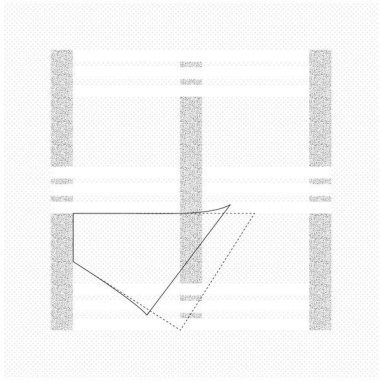
02

# LITTLE TOKYO THEATER

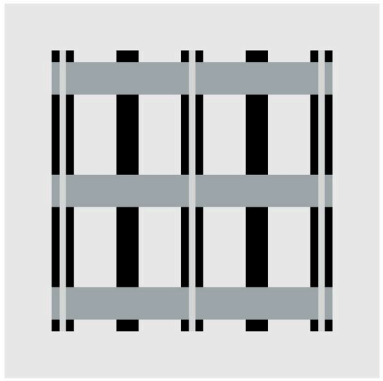
FA23, DS 1100, 1GA STUDIO
ADVISOR · MATTHEW AU

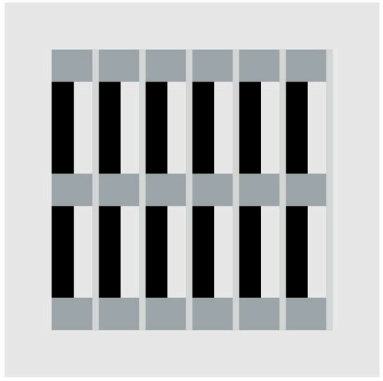
This first semester studio project stems from a study of grid systems, materiality, galleries, and theatres. The initial measurement grids were carefully constructed, using the limitations of the tape itself and the gap in named measurements vs. actual measurements. The grids operate in a way that looks at material falsities, as well as transparency. Through these measurement systems, they are then expanded into large boards. These boards are then used to produce a massing model. Ultimately realized as a performing arts theater in the Little Tokyo District, the Measurement Grid is transformed into a space that offers a place for community to gather in an unconventional theater setting where actors and patrons are active participants in the shows on display.

## GRID STUDIES

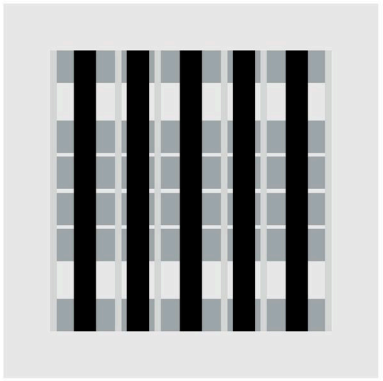


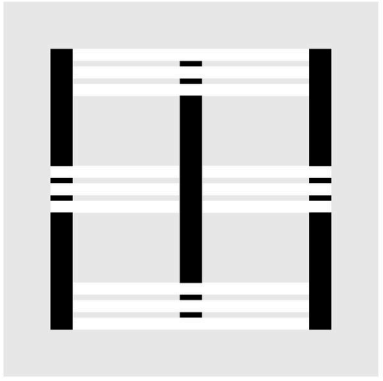
MEASUREMENT GRID DIAGRAM WITH TETRAHEDRON





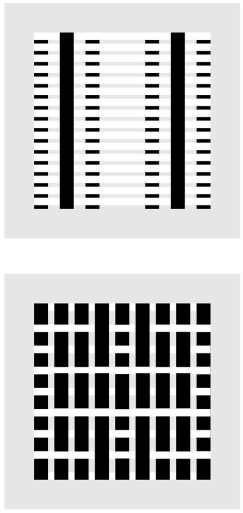
DIGITAL GRID FIGURE STUDY 1 AND 2



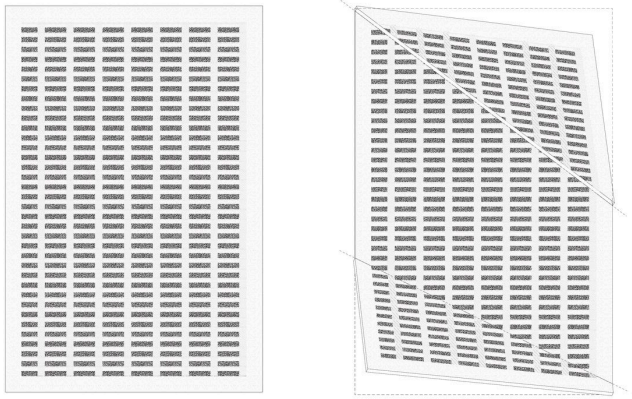


DIGITAL GRID FIGURE STUDY 3 AND 4

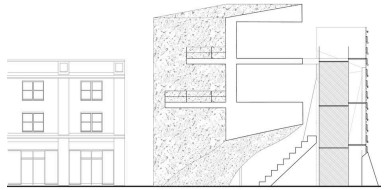
## GRID STUDIES



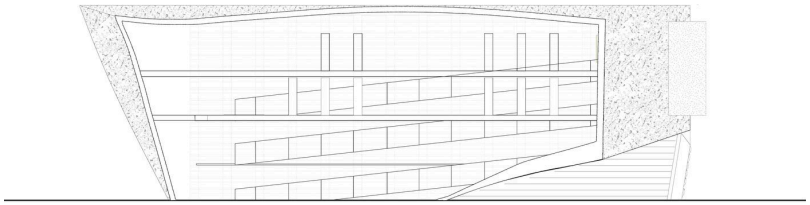
DIGITAL GRID FIGURE STUDY 5 AND 6



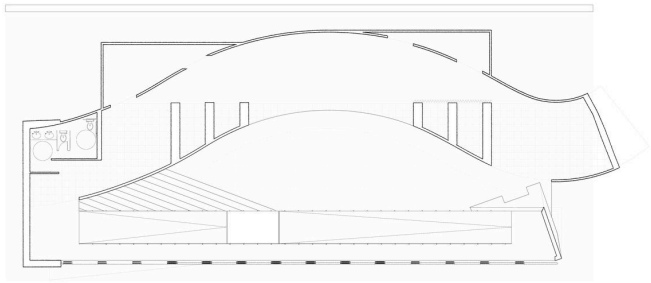
DIGITAL GRID STUDY, PRINTED BOARD FLAT AND BENT



GRID STUDY DRAWING, SHORT SECTION

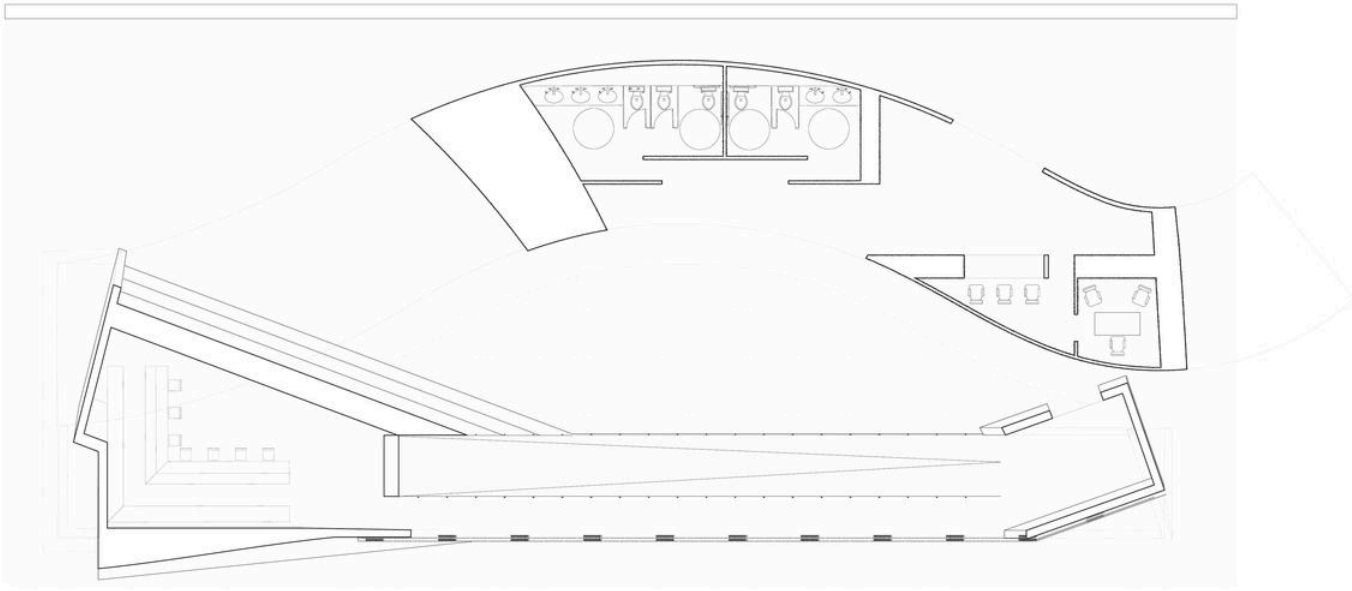


GRID STUDY DRAWING, LONG SECTION



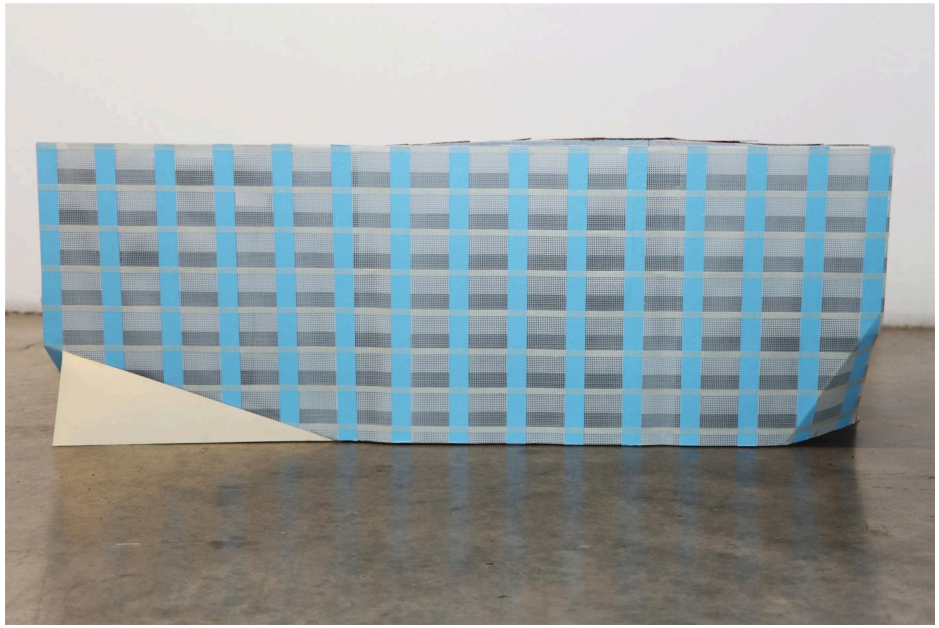
GRID STUDY DRAWING, SECTION

## GRID STUDIES

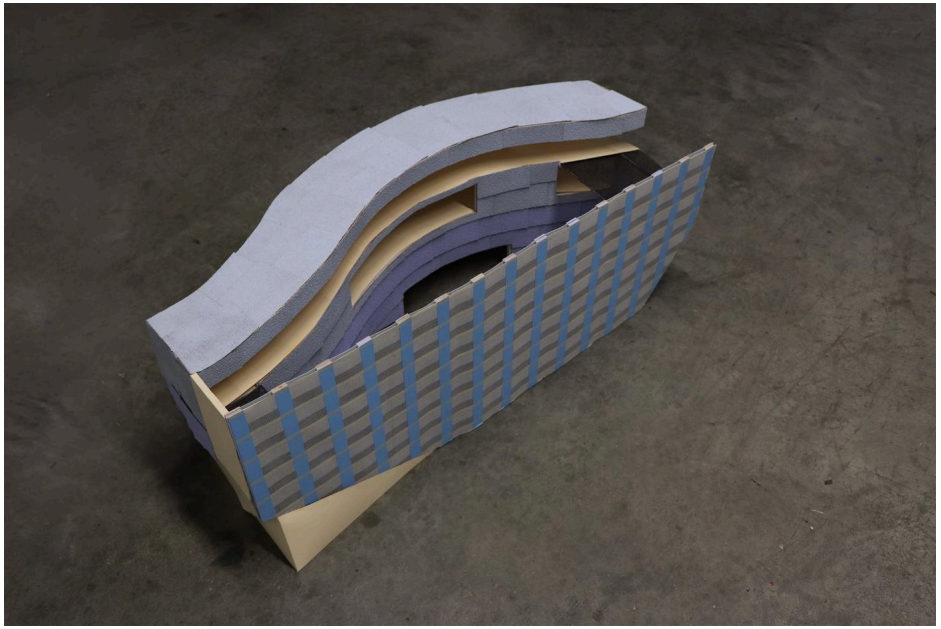


GRID STUDY DRAWING, PLAN

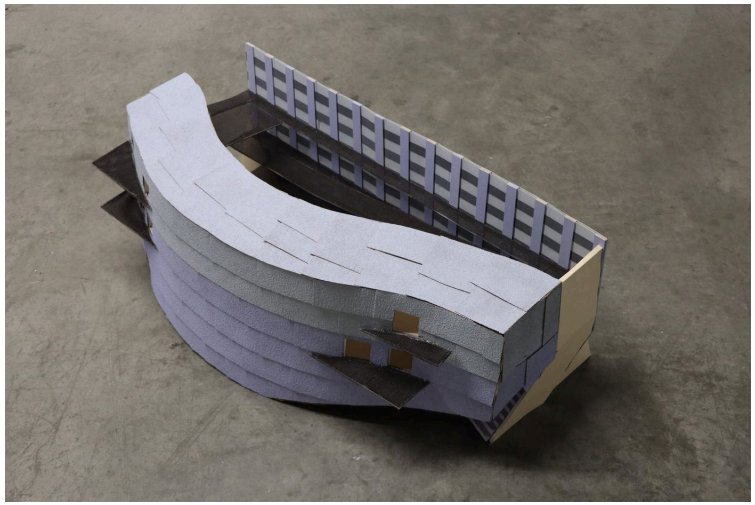
## FINAL MODEL



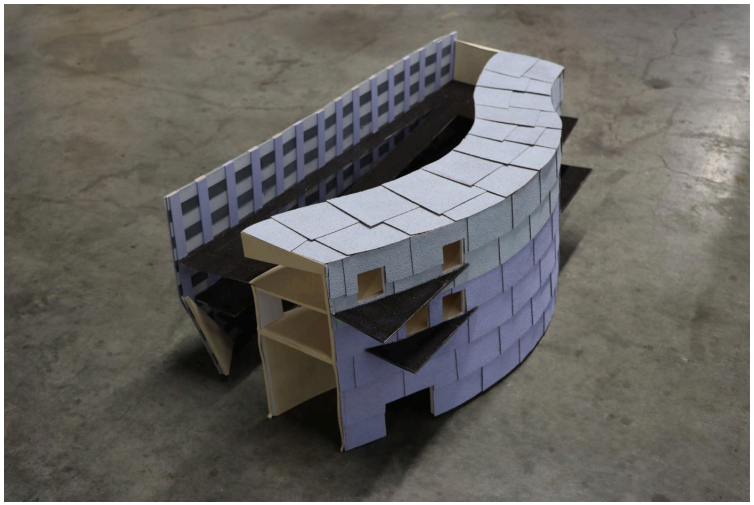
FINAL MASSING MODEL, PRIMARY VIEW



FINAL MASSING MODEL, VIEW 2



FINAL MASSING MODEL, VIEW 3

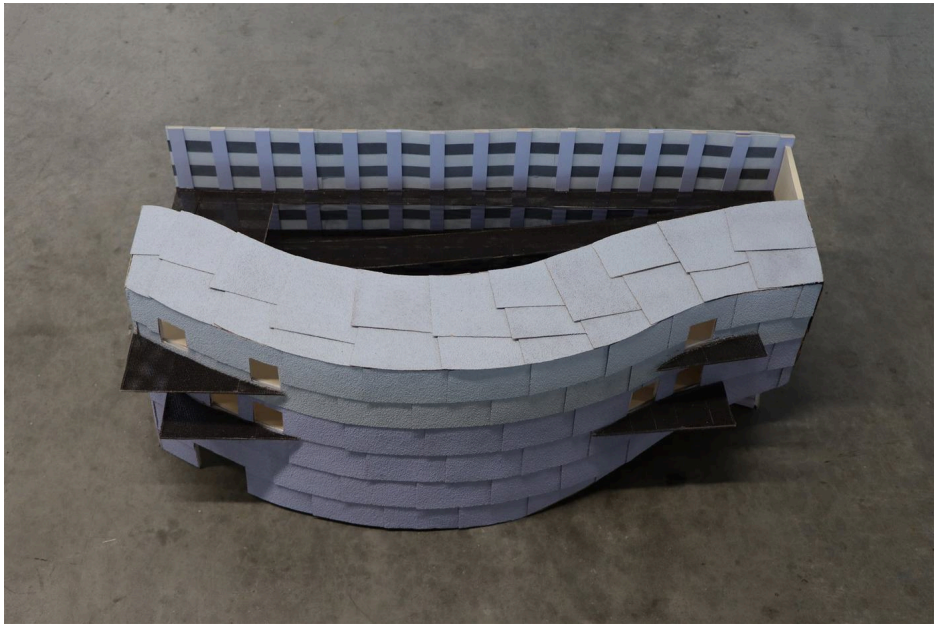


FINAL MASSING MODEL, VIEW 4

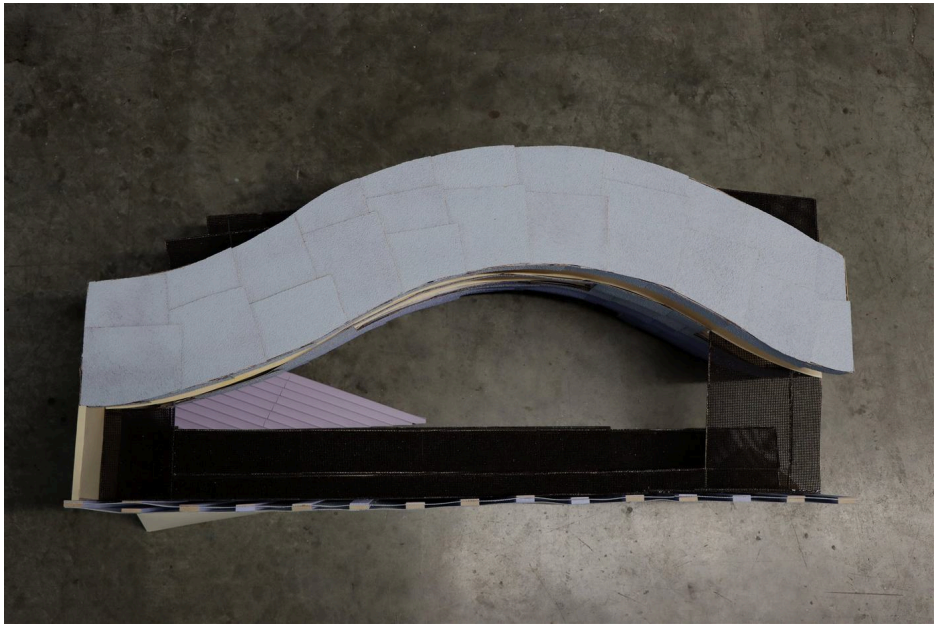


FINAL MASSING MODEL, VIEW 5

## FINAL MODEL

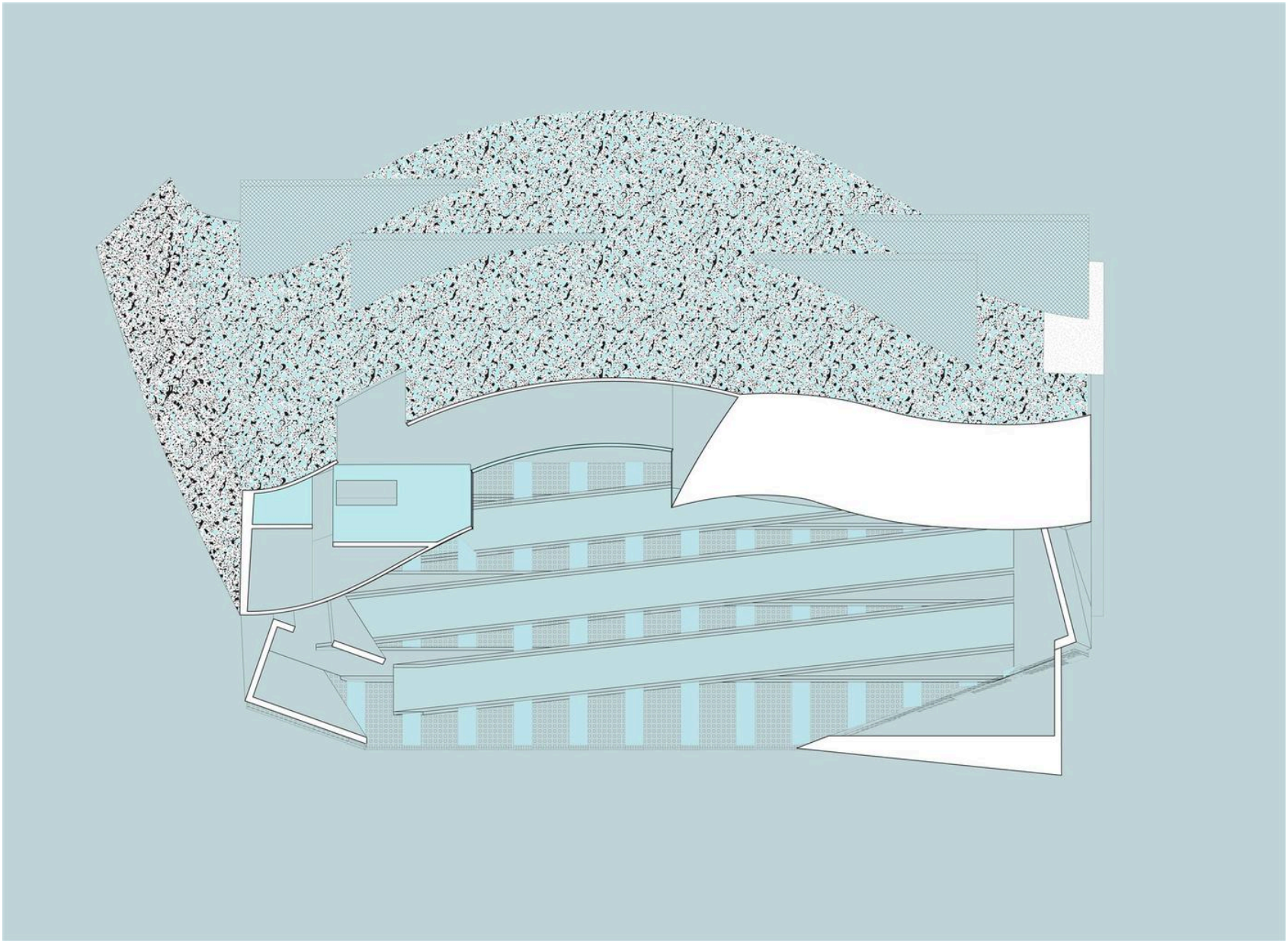


FINAL MASSING MODEL, ELEVATION VIEW

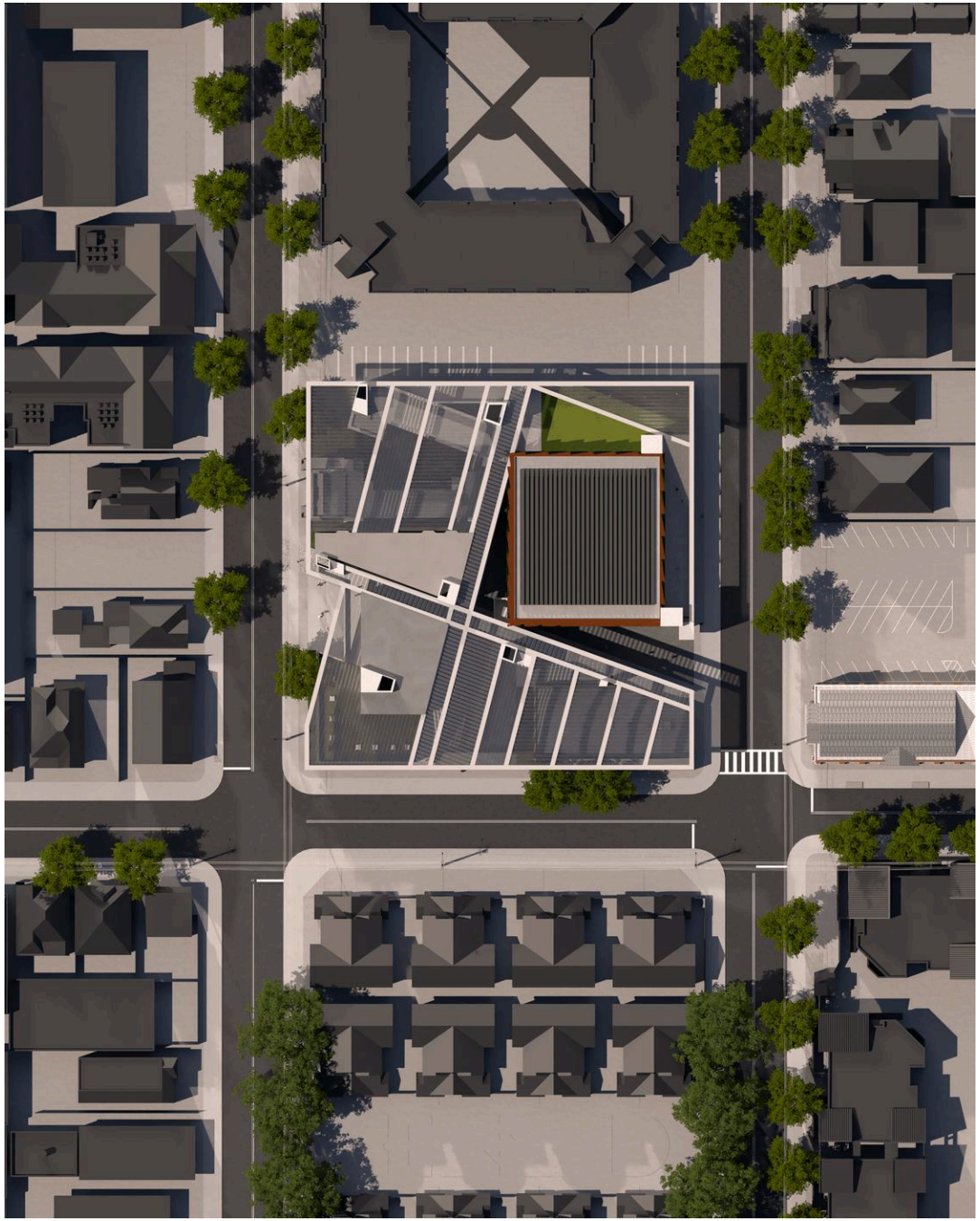


FINAL MASSING MODEL, VIEW 7

## FINAL MODEL



FINAL MODEL COMPOSITE SECTION DRAWING



03

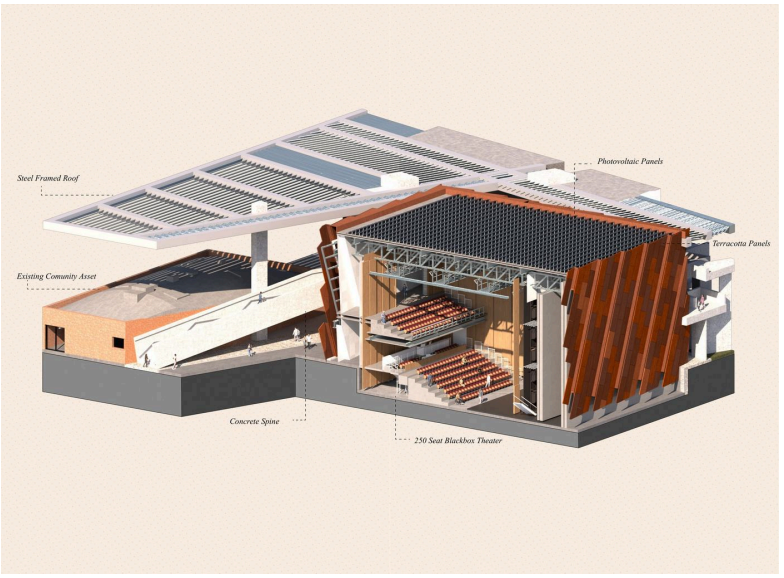
# LA DANCE SHED

FA24, 2GA STUDIO, WITH MILAN SLEDGE
ADVISOR · MARCELO SPINA

This project for the Los Angeles Dance Project (LADP) seeks to bring the community of Pico-Union together on a campus that serves and connects the community. Located directly across from the Pico-Union Project, the site was surgically opened up in order to connect via sightlines and circulation to this important community asset. A unique goal among this project was to preserve many of the existing structures on the site and repurposing the materials from buildings that were to be removed. With a raised circulatory system, visitors are encouraged to meander throughout the site, engaging with it from within the in-between spaces. The roof serves as not only shade for warm summer days, but also acts as an organizing device for the discrete elements of the site, bringing them together in a collective, just as is intended with the surrounding community.



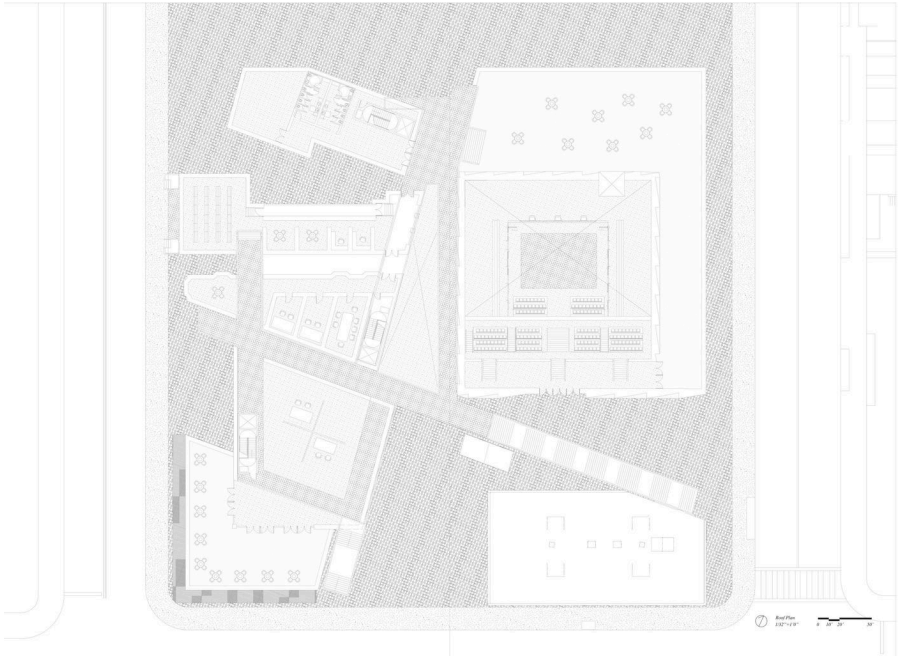
RENDER | FROM THE PICO-UNION SIDE OF THE SITE



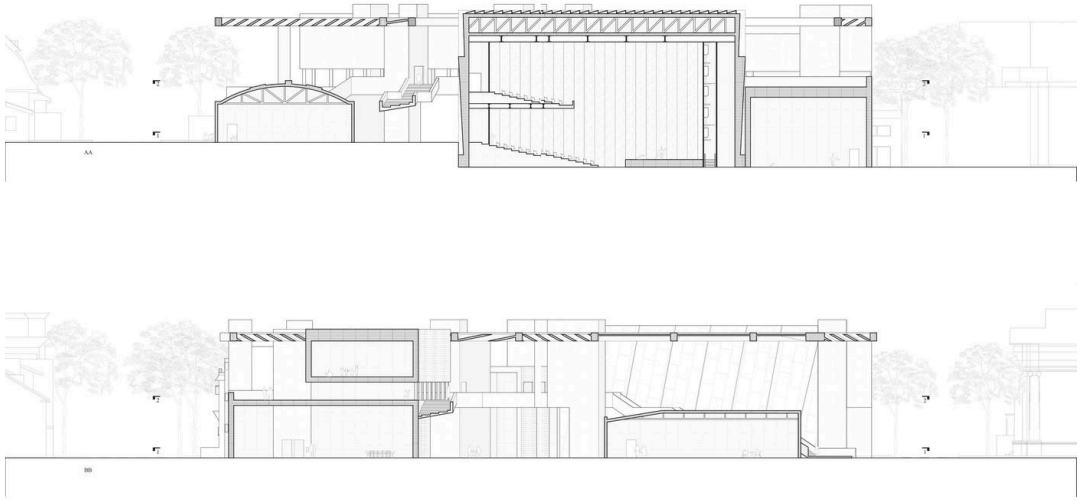

AXONOMETRIC | STEEL-FRAMED ROOF, CONCRETE SPINE AND BLACK BOX THEATER



PLAN | LEVEL 1



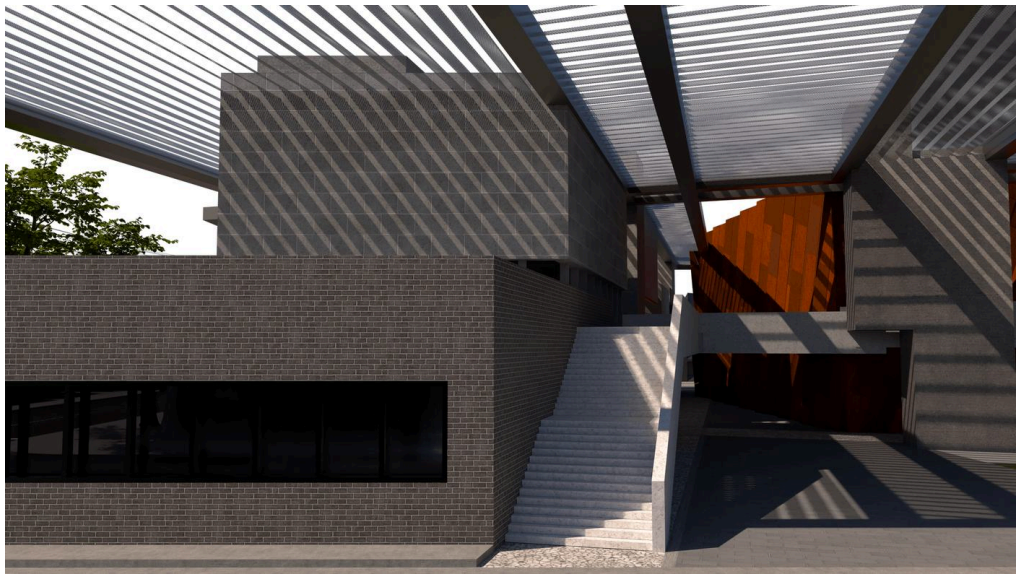
PLAN | LEVEL 2



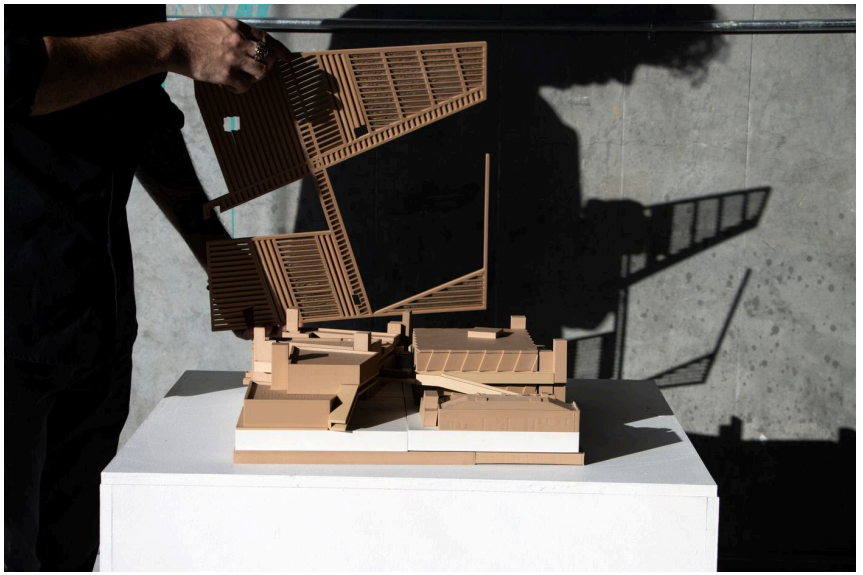
SECTIONS | THROUGH THE THEATER AND SPINE



ELEVATIONS | STREET FRONTAGES, TERRACOTTA PANEL FACADE



RENDER | TOWARD THE CAMPUS CAFE



MODEL | THE ROOF LIFTED OFF THE CAMPUS BELOW



MODEL | SETTING THE ROOF PLATES OVER THE SITE